

Windows Server Patch Management Lab using WSUS

Build a lab where a Windows Server manages updates for client machines using WSUS and Group Policy.

Objectives

- Deploy and configure Windows Server Update Services (WSUS)
- Configure Active Directory and Group Policy for update management
- Create computer groups for phased deployment
- Synchronize updates from Microsoft Update
- Approve and deploy updates to managed devices
- Monitor update compliance and reporting
- Verify successful patch installation on client systems

Skills Demonstrated

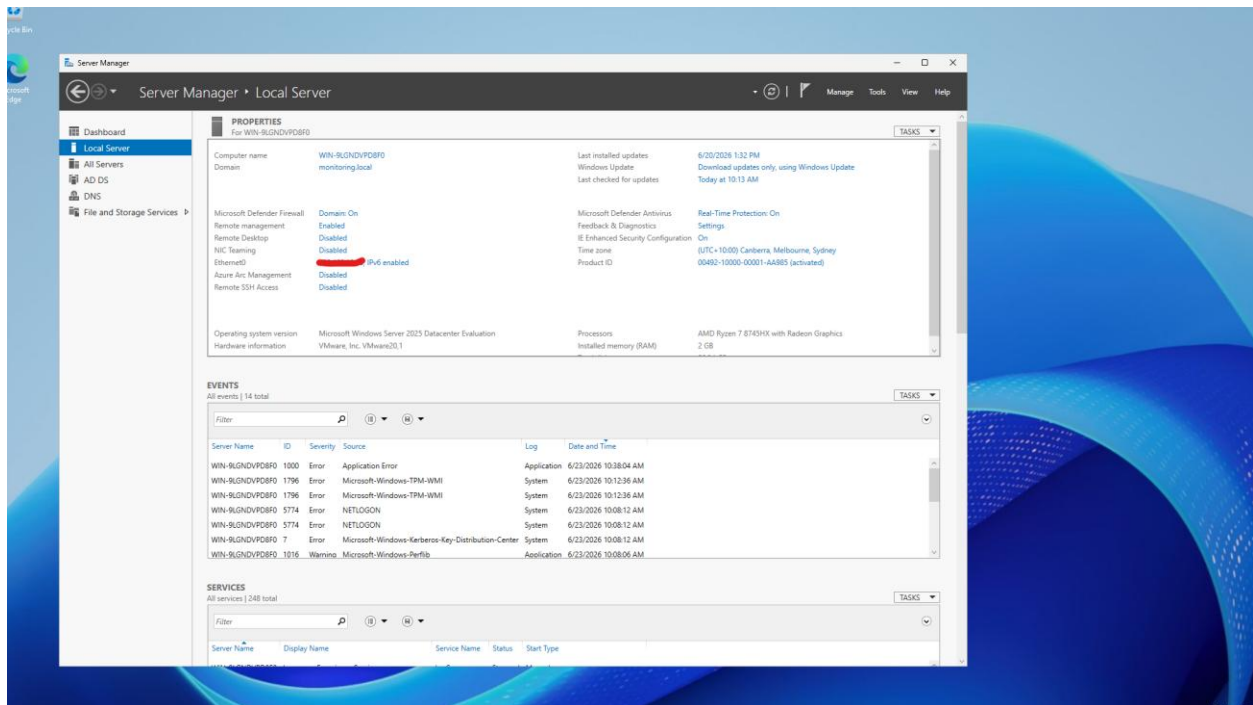
- Windows Server Administration
- Active Directory Administration
- DNS Management
- WSUS Administration
- Group Policy Management
- Patch Management
- Windows Endpoint Management
- PowerShell
- System Administration
- Troubleshooting and Compliance Reporting

Lab Requirements

You need:

1. DC – Windows Server Domain Controller
2. WSUS – Windows Server for WSUS
3. CLIENT – Windows 10/11 Pro client

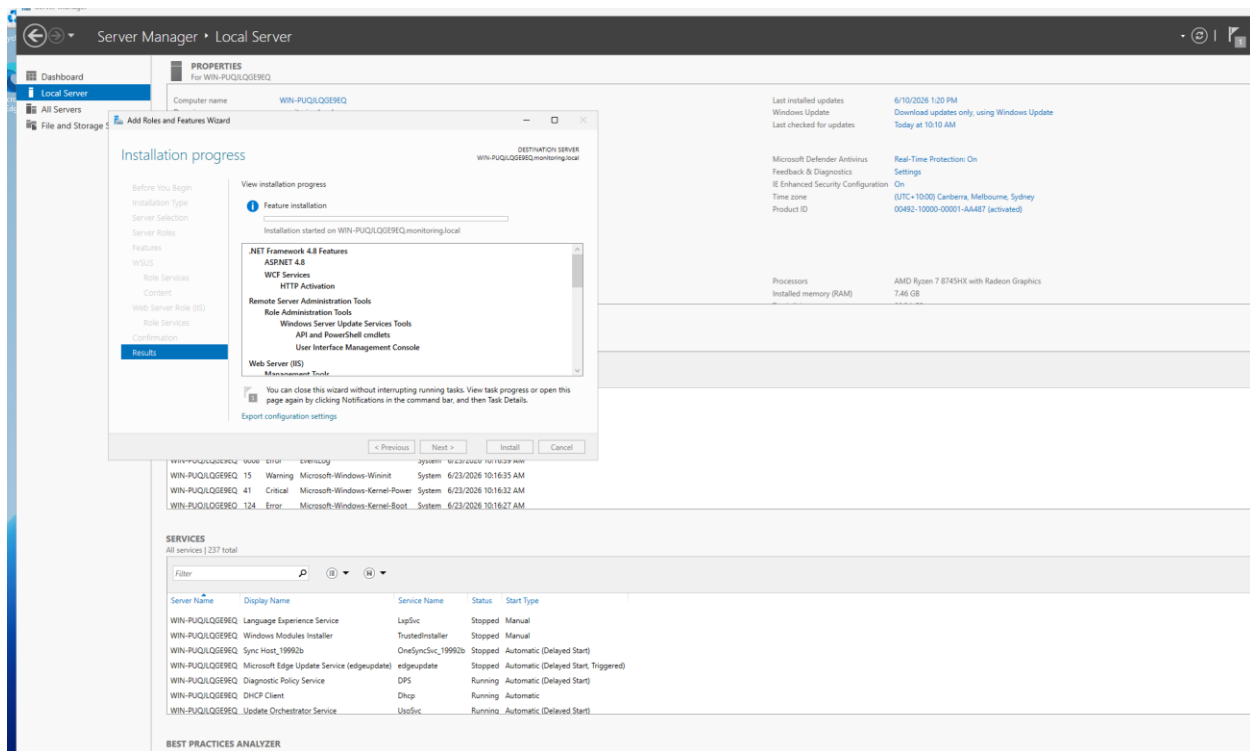
Windows server 2025 has been created and promoted into domain controller



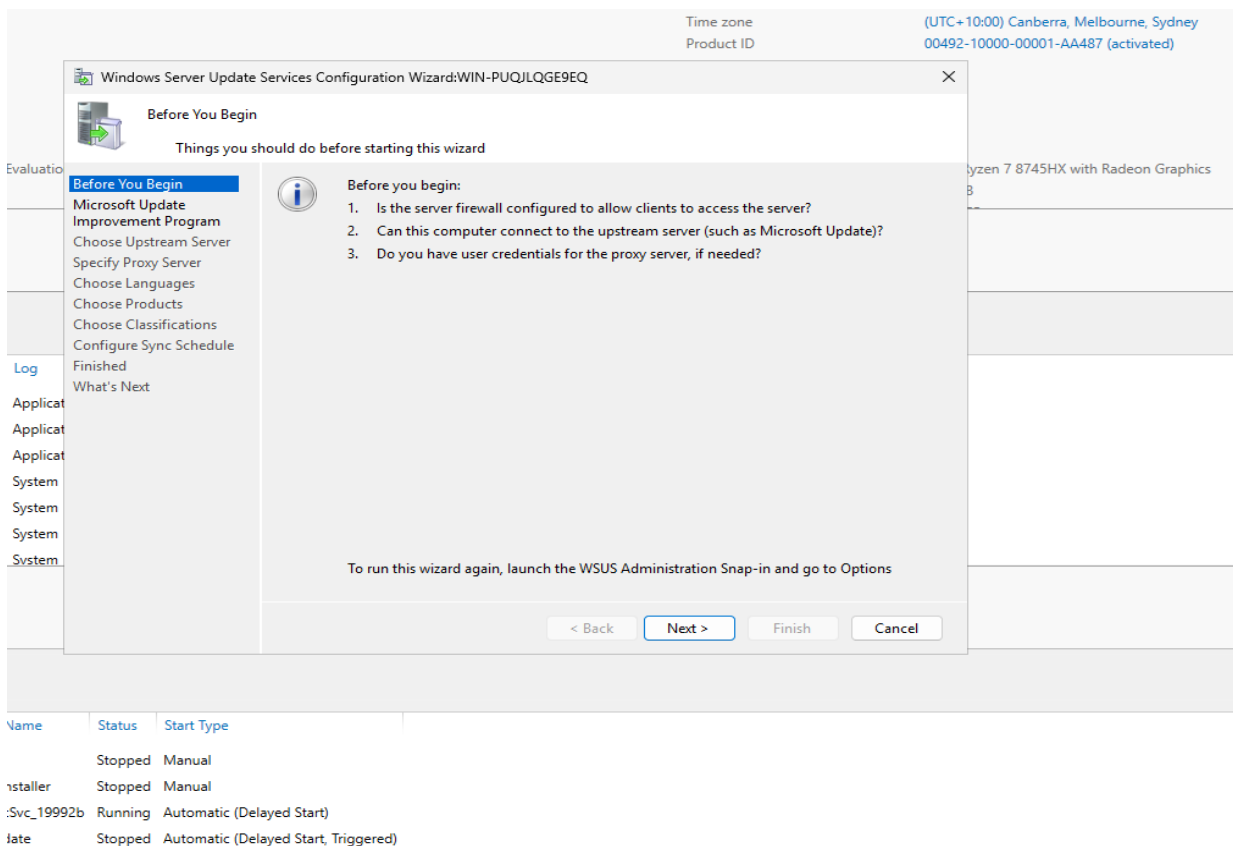
Create WSUS Server and join to the domain

- Open Server Manager.
- Go to Manage > Add Roles and Features
- Select Windows Server Update Services

And Complete installation.



Open Tools > Windows Server Update Services. And Run WSUS post-install configuration.



Configure WSUS

In WSUS Configuration Wizard:

1. Synchronize from Microsoft Update.
2. Choose language: English only.
3. Products:
 - Windows 10
 - Windows 11
 - Windows Server 2022
 - Microsoft Defender Antivirus
4. Classifications:
 - Critical Updates
 - Security Updates
 - Update Rollups
 - Definition Updates
5. Set sync schedule:

Daily at 2:00 AM
6. Start initial synchronization.

IE Enhanced Security Configuration On
Time zone (UTC+10:00) Canberra, Melbourne
Product ID 00492-10000-00001-AA487 (activ

Windows Server Update Services Configuration Wizard:WIN-PUQJLQGE9EQ

Connect to Upstream Server
Download update information from Microsoft Update

Before You Begin
Microsoft Update Improvement Program
Choose Upstream Server
Specify Proxy Server
Choose Languages
Choose Products
Choose Classifications
Configure Sync Schedule
Finished
What's Next

To configure Windows Server Update Services on the following screens, we need to apply your upstream server and proxy server settings and synchronize information about available updates.

Information to be downloaded includes:

Types of updates available
Products that can be updated
Available languages

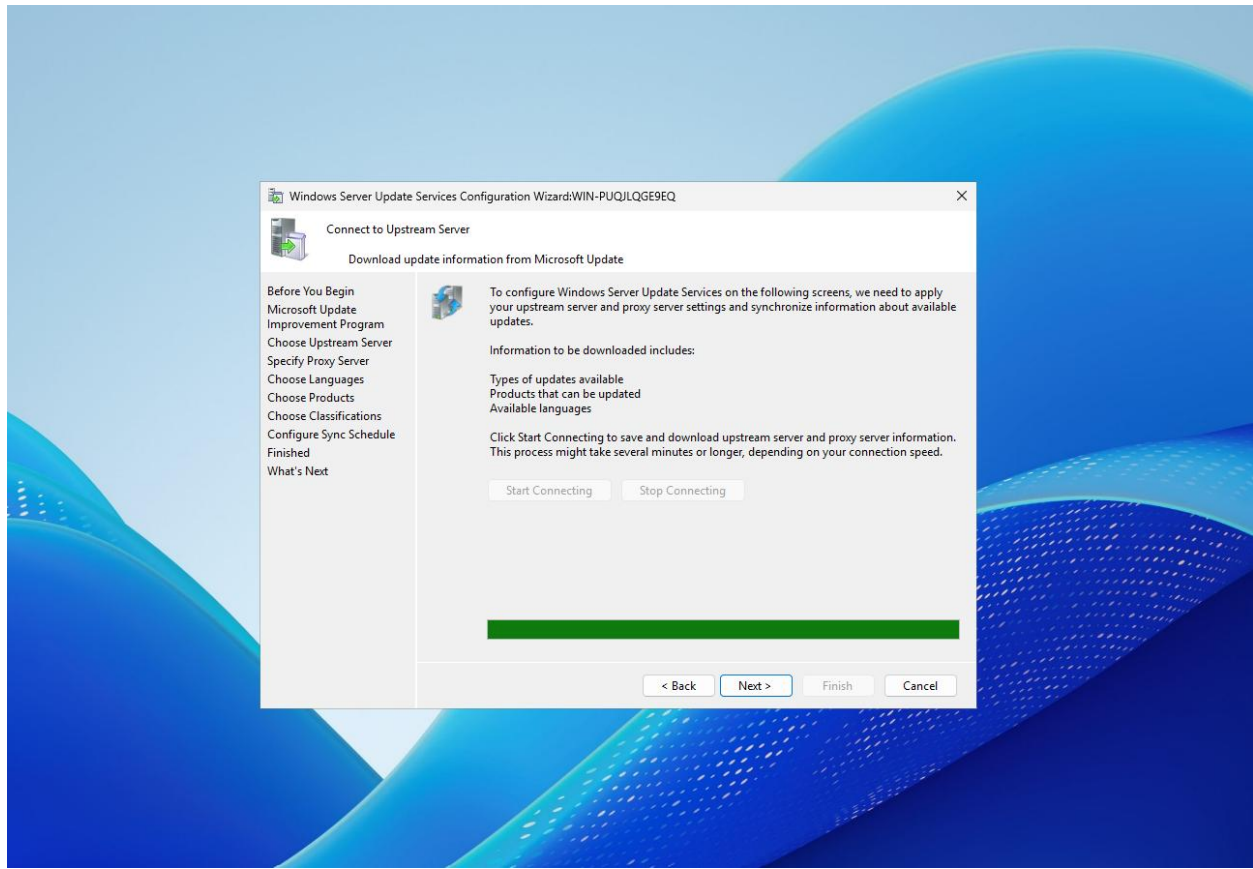
Click Start Connecting to save and download upstream server and proxy server information. This process might take several minutes or longer, depending on your connection speed.

Start Connecting Stop Connecting

< Back Next > Finish Cancel

Service Name	Status	Start Type
ipSvc	Stopped	Manual
ustedInstaller	Stopped	Manual
neSyncSvc_19992b	Running	Automatic (Delayed Start)
lgeupdate	Stopped	Automatic (Delayed Start, Triggered)

Synchronization is completed



Choose products

Expand Microsoft Corporation and select:

- Windows 11
- Windows Server

That's all you need.

For , Windows 10 client we need to select:

- Windows 10, version 1903 and later

Feedback & Diagnostics

IE Enhanced Security Configuration

Time zone

Product ID

Settings

On

(UTC+10:00) Canberra, Melbourne, Sydney

00492-10000-00001-AA487 (activated)

Windows Server Update Services Configuration Wizard:WIN-PUQLQGE9EQ

Choose Products

Select the Microsoft products to be updated

Before You Begin

Microsoft Update Improvement Program

Choose Upstream Server

Specify Proxy Server

Choose Languages

Choose Products

Choose Classifications

Configure Sync Schedule

Finished

What's Next

You can specify the products for which you want updates.

Products:

☐ Azure Stack HCI

☐ EU Browser Choice Update-For Europe Only

☐ Microsoft Defender Antivirus

☐ Microsoft Defender for Endpoint

☐ Microsoft Defender for Endpoint

☒ Microsoft Edge

☒ Microsoft Server operating system-21H2

☐ Microsoft Server Operating System-22H2

☐ Microsoft Server Operating System-23H2

☐ Microsoft Server Operating System-24H2

☐ OOBE ZDP

☐ Server 2022 Hotpatch Category

☐ Surface Hub 2S drivers

All products, including products that are added in the future.

< Back

Next >

Finish

Cancel

Service Name

Status

Start Type

LxpSvc

Stopped

Manual

TrustedInstaller

Stopped

Manual

OneSyncSvc_19992b

Running

Automatic (Delayed Start)

edgeupdate

Stopped

Automatic (Delayed Start, Triggered)

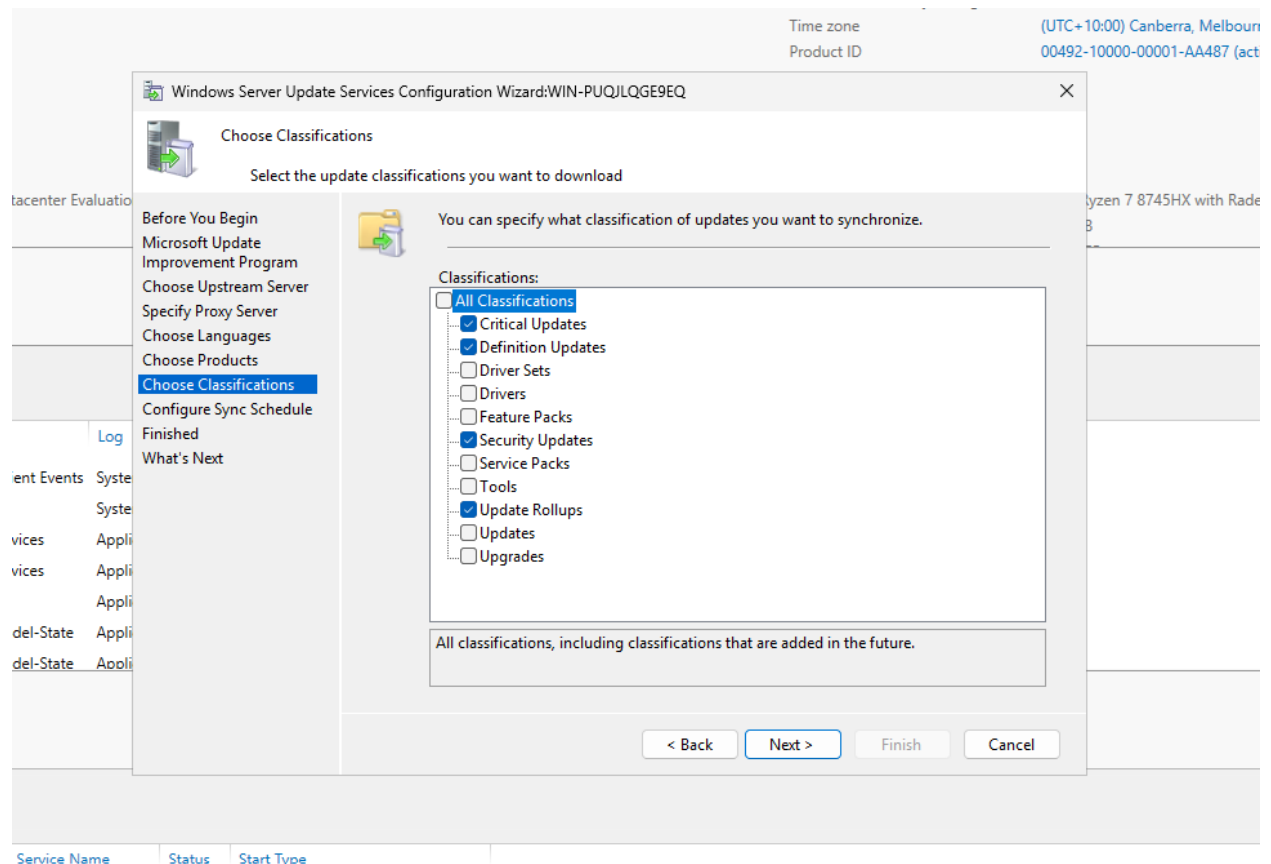
DPS

Running

Automatic (Delayed Start)

Server Manager

Choose the type of update



Set sync schedule:

Daily at 2:00 AM

Microsoft Defender Antivirus
Feedback & Diagnostics
IE Enhanced Security Configuration
Time zone
Product ID

Real-Time Protection: On
Settings
On
(UTC+10:00) Canberra, Melbourne, Sydne
00492-10000-00001-AA487 (activated)

2025 Datacenter Evaluation

Log

:-DNS Client Events System
:-date Services Application
:-date Services Application
:-T##WID Application
:-AppModel-State Application
:-AppModel-State Application

Windows Server Update Services Configuration Wizard:WIN-PUQJLQGE9EQ

Finished
Finished initial configuration of your server

Before You Begin
Microsoft Update Improvement Program
Choose Upstream Server
Specify Proxy Server
Choose Languages
Choose Products
Choose Classifications
Configure Sync Schedule
Finished
What's Next

✓

You have now finished configuring your WSUS server. You can go on to launch the WSUS Administration Console or start the initial synchronization.

☐ Launch the Windows Server Update Services Administration Console
Get started administering your organization by starting the Windows Server Update Services Administration Console.

☐ Begin initial synchronization
The initial synchronization with Microsoft Update or your upstream server can take some time. The length of the initial synchronization depends on the number of products, update classifications, and languages you have selected, as well as the speed of your Internet connection.

< Back

Next >

Finish

Cancel

Service Name

Status

Start Type

LxpSvc

Stopped

Manual

TrustedInstaller

Stopped

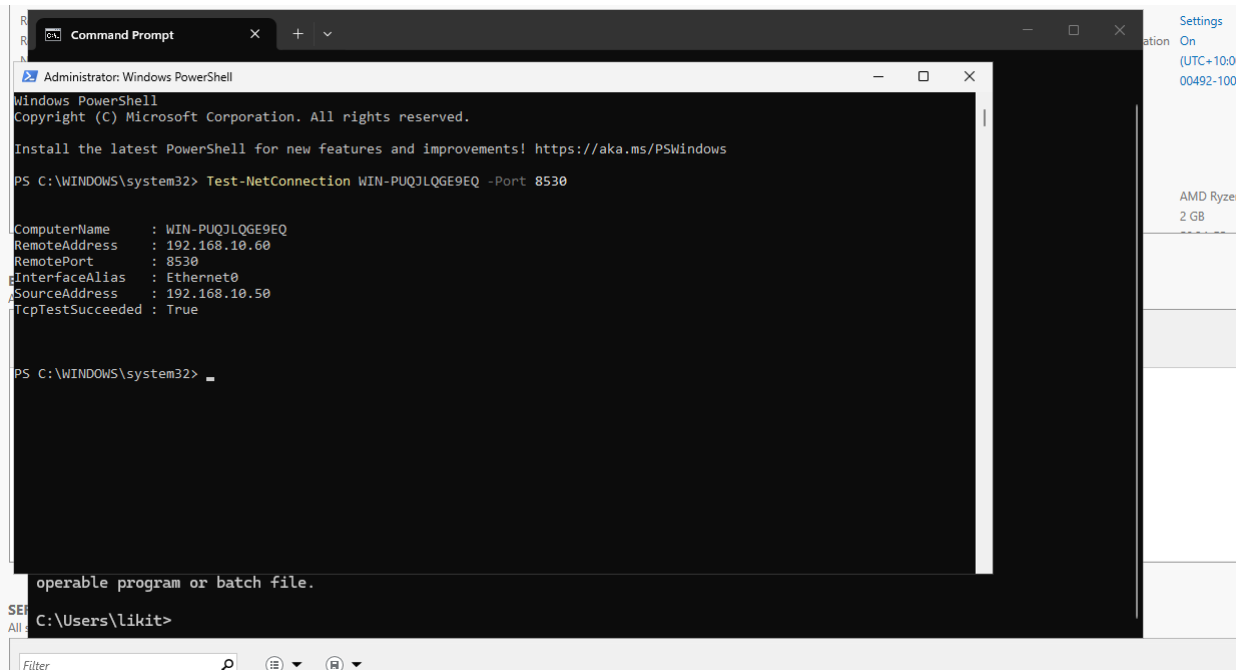
Manual

OneSyncSvc_19992b

Running

Automatic (Delayed Start)

Test whether WSUS is listening on port 8530



Create Computer Groups in WSUS server:

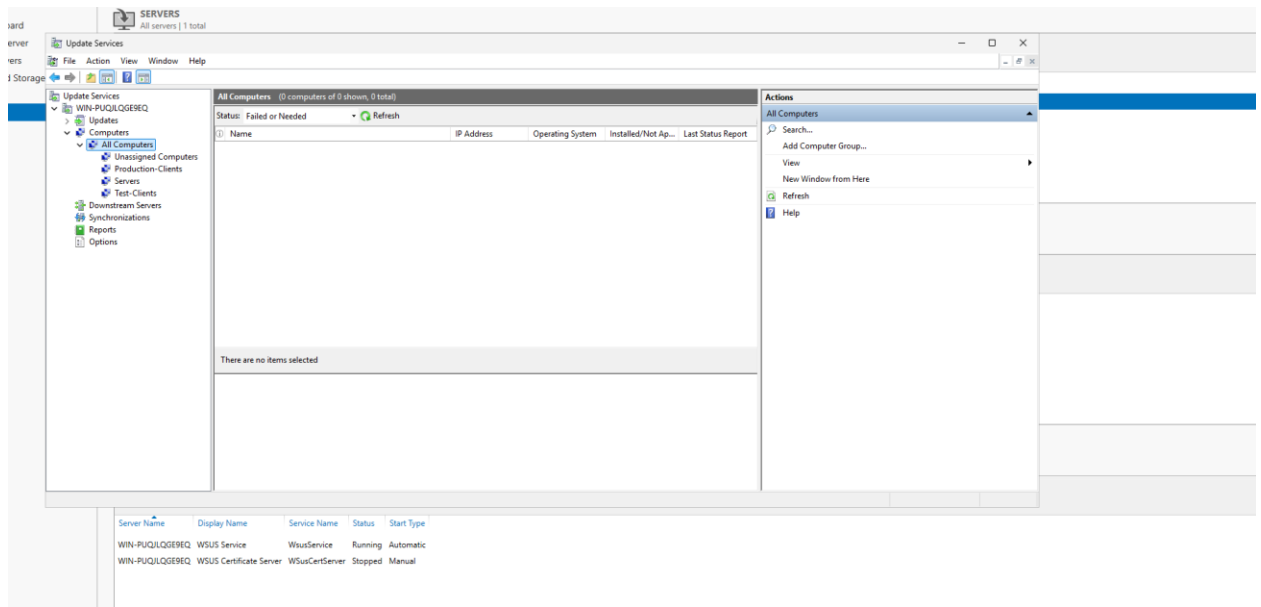
On **WSUS**:

1. Open **Windows Server Update Services**
2. Expand:

WSUS server

└ Computers

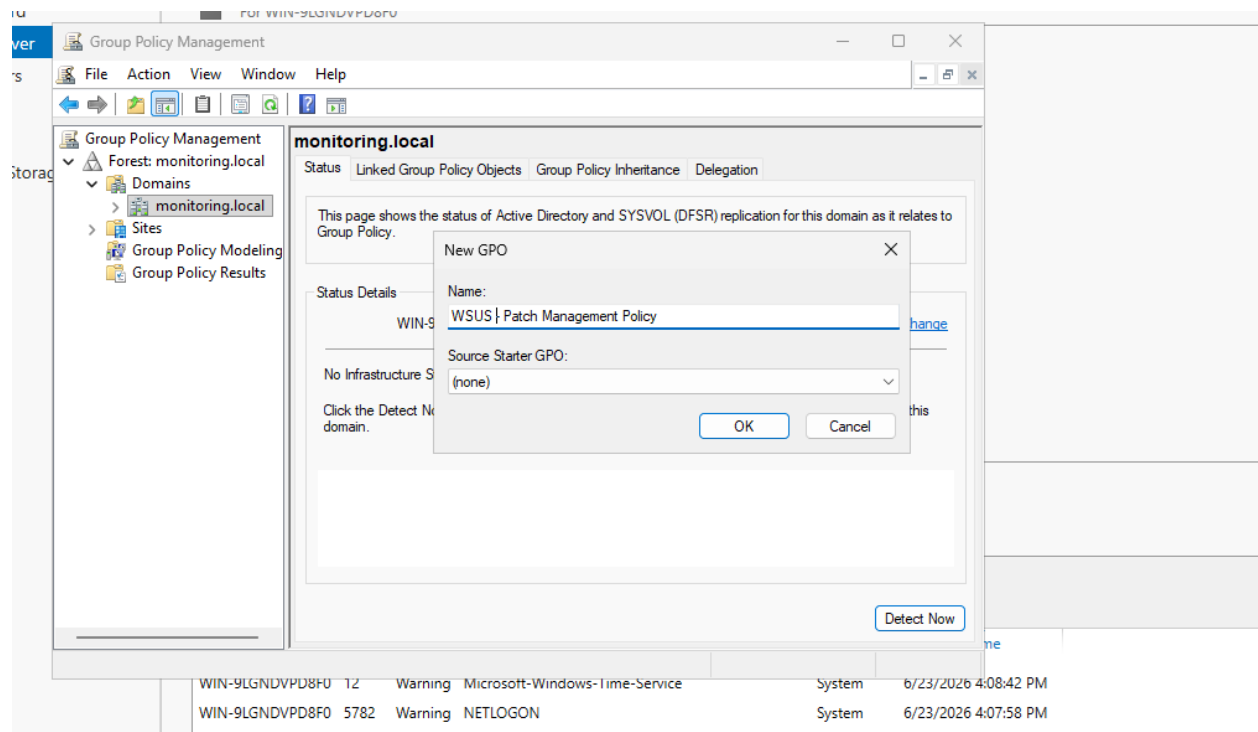
3. Right-click **Computers**
4. Select **Add Computer Group**
5. Create these groups:
 - Test-Clients
 - Servers
 - Production-Clients



Create Group Policy: To create GPO for the client computers, go to domain controller

On DC01: Open: Group Policy Management

Create a new GPO: name it as WSUS - Patch Management Policy

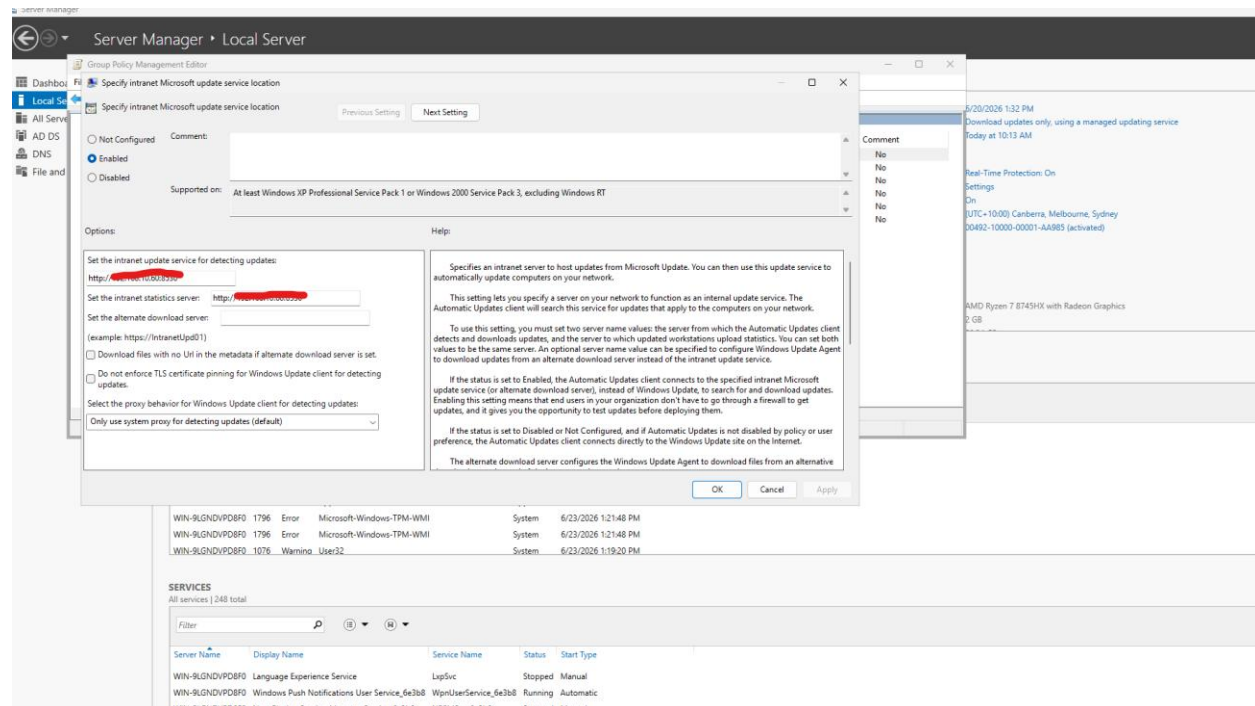


1. Specify intranet Microsoft update service location

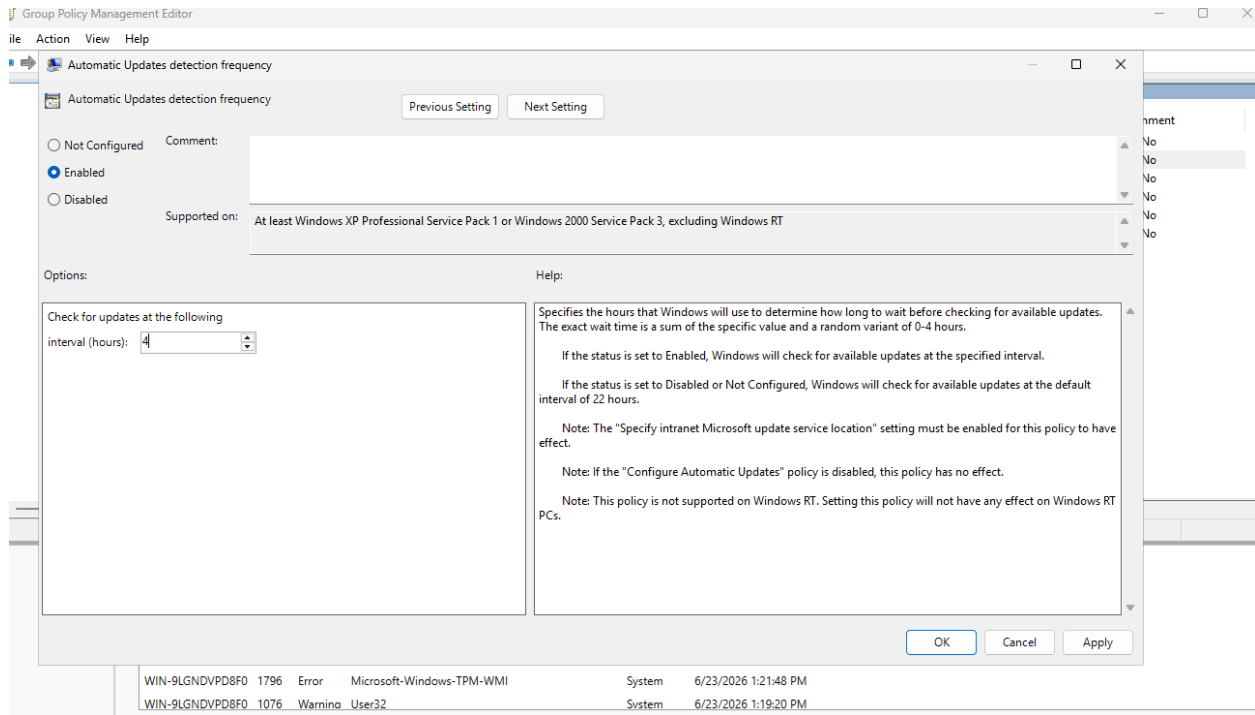
Set to Enabled

For both fields enter:

http://WSUS01:8530

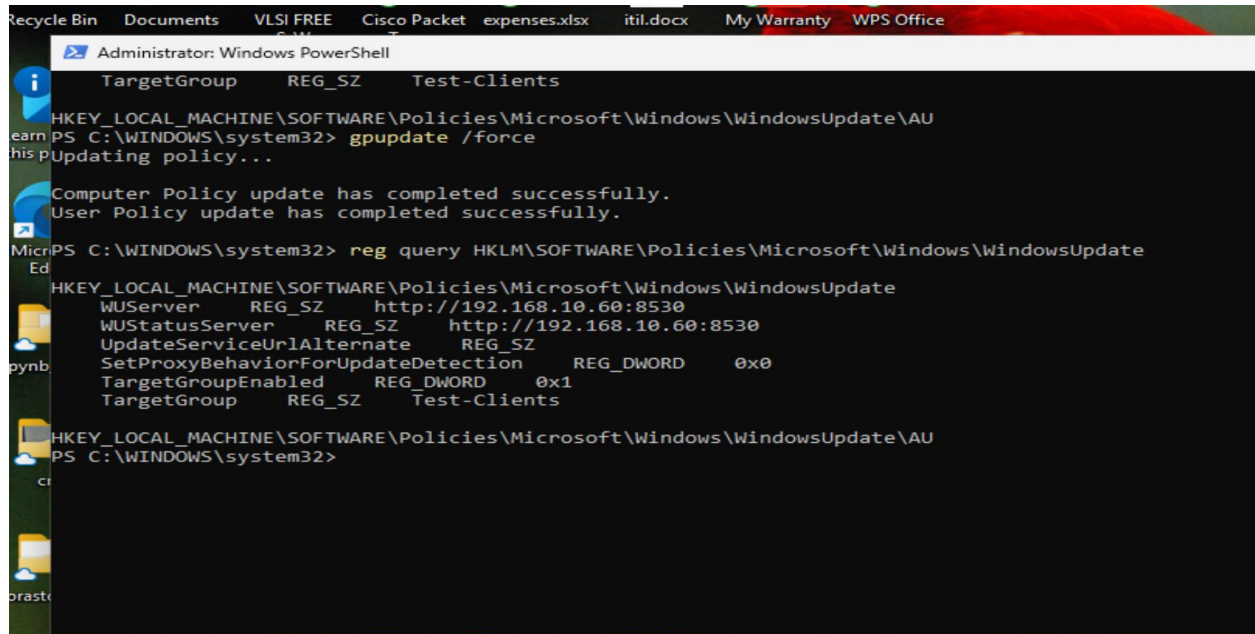


Automatic update every 4 hours

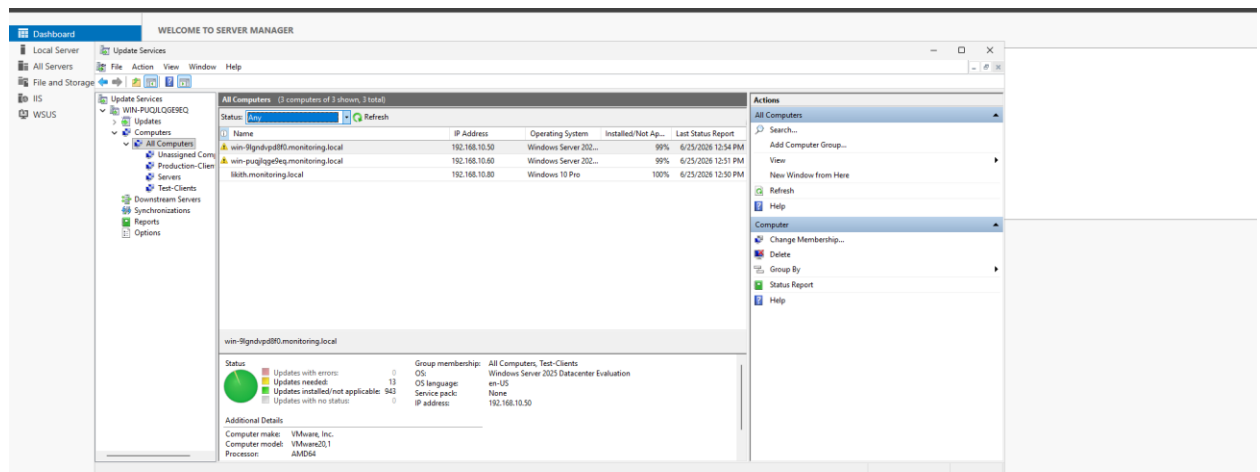
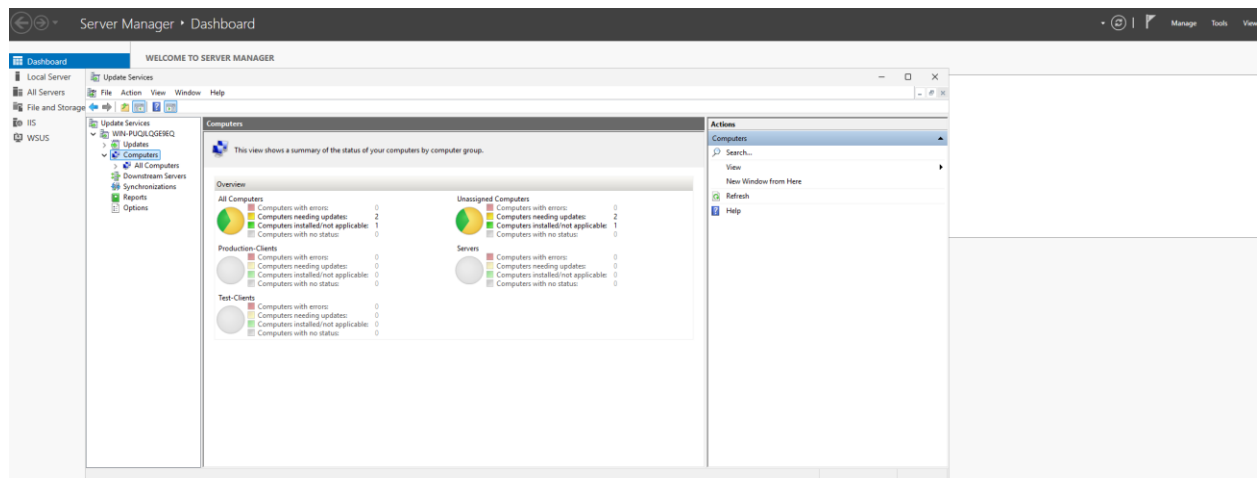


Force Policy on Client:

Run gpupdate /force on windows 11 computer



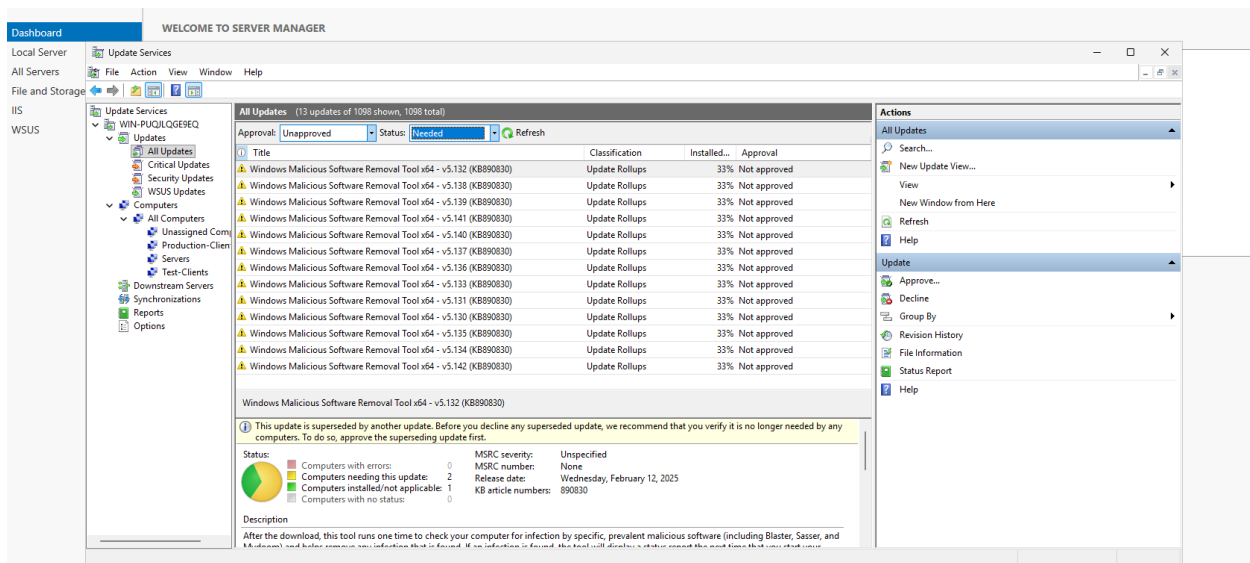
Confirm Client Appears in WSUS



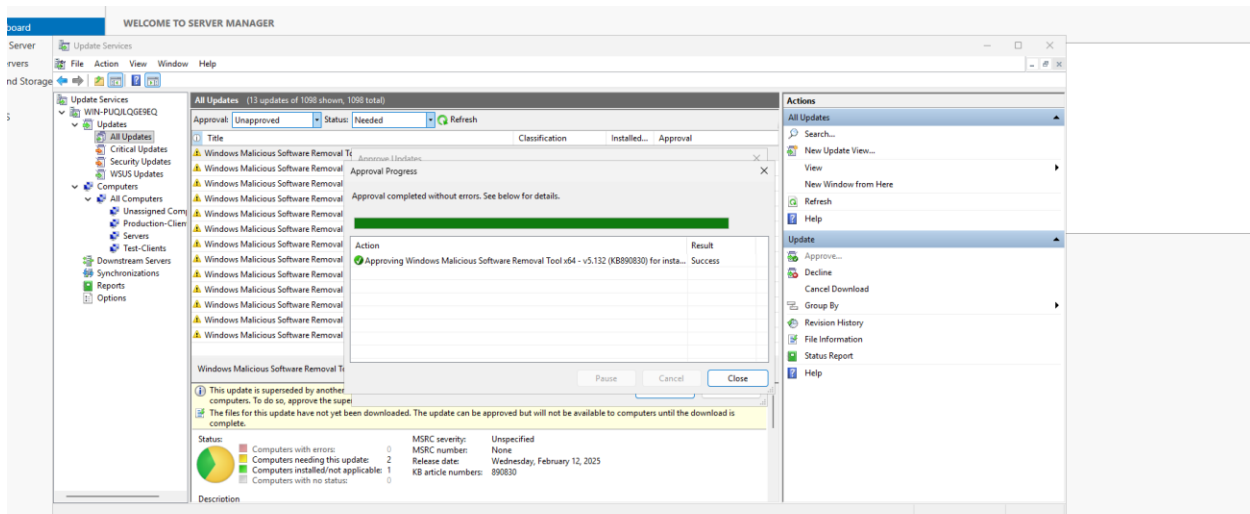
The screenshot shows last status report received client is showing 13 updates needed

Approve Updates

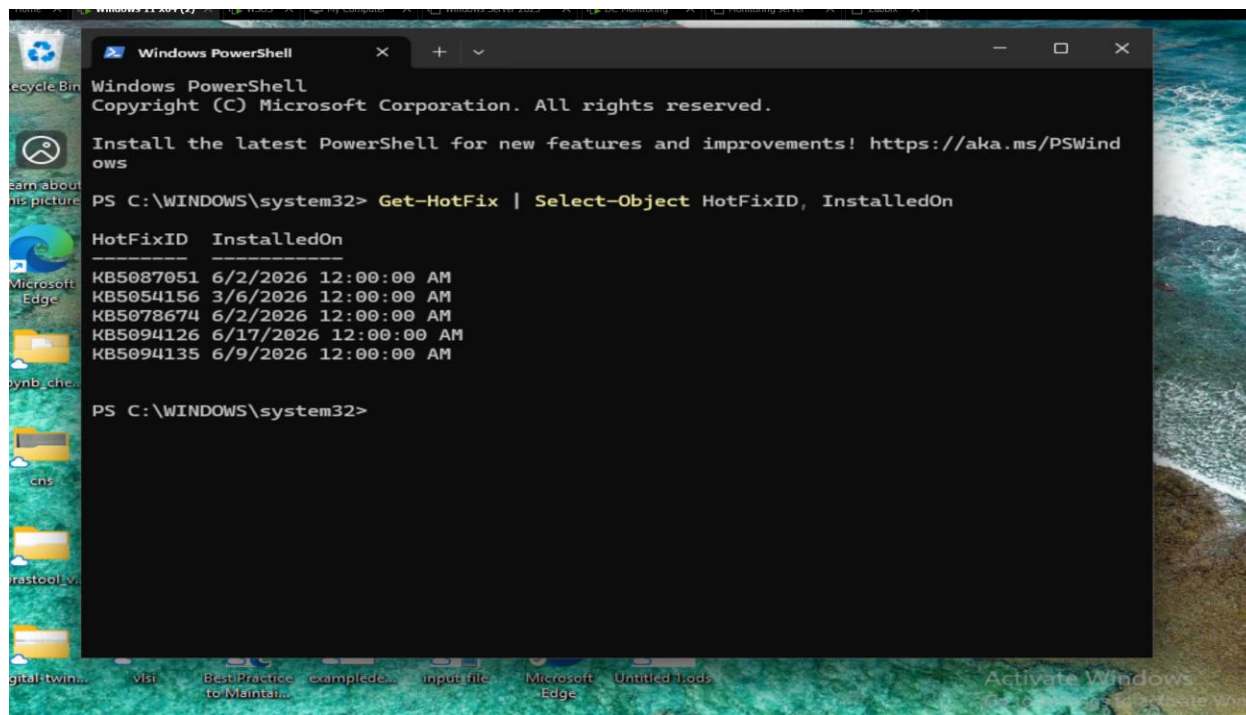
Go to: Updates → All Updates



Click on Approve to update the updates



If approved updates install, WSUS deployment is working end-to-end



Validation

WSUS successfully verified:

- Client registration
- Group Policy application
- Update synchronization
- Update approval workflow
- Compliance reporting

PowerShell Verification

Command used:

Get-HotFix | Select-Object HotFixID, InstalledOn

Installed updates detected:

- KB5087051
- KB5054156
- KB5078674
- KB5094126

- KB5094135

Results:

- Successfully deployed a centralized patch management solution.
- Configured WSUS to manage Windows updates across domain-joined systems.
- Implemented Group Policy-based update management.
- Successfully approved and deployed updates to client devices.
- Verified patch compliance through WSUS reporting and PowerShell validation.
- Demonstrated enterprise patch lifecycle processes including synchronization, approval, deployment, monitoring, and compliance reporting.